

JARNASKLUBB, Sweden

WMO: 022840

Lat: 63.435N

Lon: 19.673E

Elev: 19

StdP: 14.69

Time Zone: 1.00 (EUC)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-3.4	1.3	-8.5	3.5	-3.6	-3.7	4.5	1.6	35.4	32.0	32.9	31.0	8.5	350	0.722

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.8	72.3	65.3	69.7	63.4	67.2	61.2	67.1	71.2	64.6	68.2	62.5	66.0	9.1	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.3	93.7	70.1	62.9	85.9	67.2	60.9	79.8	64.7	31.6	71.4	29.7	68.4	28.2	66.1	73.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 32.0	27.5	23.4	DB	-8.0	75.8	7.5	2.8	-13.4	77.8	-17.8	79.5	-22.0	81.0	-27.5	83.1	(4)
(5)			WB	-8.7	67.3	6.9	2.3	-13.7	69.0	-17.7	70.3	-21.6	71.6	-26.6	73.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	40.4	23.9	22.1	27.8	35.6	45.2	54.3	60.9	59.0	51.6	41.4	33.7	28.0		
	DBStd	14.97	9.88	10.89	7.77	4.95	5.74	4.60	4.28	4.44	5.04	6.14	7.55	9.30		
	HDD50	4383	808	781	689	433	173	13	0	1	42	271	489	683		
	HDD65	9003	1273	1201	1154	883	614	322	142	192	403	732	939	1148		
	CDD50	878	0	0	0	0	23	142	339	281	89	4	0	0		
	CDD65	24	0	0	0	0	0	1	16	7	0	0	0	0		
	CDH74	33	0	0	0	0	1	1	27	4	0	0	0	0		
	CDH80	1	0	0	0	0	0	0	1	0	0	0	0	0		
	WSAvg	10.4	12.0	11.0	10.1	8.9	9.0	9.1	9.0	9.5	10.1	11.1	12.3	12.6		
	PrecAvg	23.3	1.6	1.2	1.1	1.3	1.5	1.9	2.7	2.7	2.3	2.3	2.4	1.8		
(15) Precipitation	PrecMax	35.6	2.9	2.4	1.8	3.3	2.9	3.1	6.9	5.7	5.8	5.8	5.9	3.1		
	PrecMin	16.9	0.4	0.3	0.4	0.2	0.2	1.0	0.4	0.3	0.8	0.6	0.6	0.6		
	PrecStd	4.0	0.7	0.6	0.4	0.8	0.6	0.7	1.6	1.6	1.3	1.3	1.3	0.8		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.3	43.7	48.0	54.2	65.7	71.0	77.3	73.8	64.5	55.2	49.2	43.7		
		MCWB	37.6	38.4	40.4	45.1	55.1	60.6	68.9	67.0	56.1	50.7	45.5	40.7		
	2%	DB	38.3	39.6	43.6	48.8	60.6	66.9	73.5	71.0	62.2	52.9	46.3	41.6		
		MCWB	36.1	35.9	37.5	41.7	51.8	57.3	66.2	65.3	56.7	49.6	44.7	39.7		
	5%	DB	36.9	36.8	39.8	45.6	57.2	64.2	71.0	68.7	60.5	51.7	44.7	40.1		
		MCWB	35.1	34.1	35.2	39.8	49.4	56.4	64.5	62.9	56.3	49.0	43.3	38.4		
	10%	DB	35.5	34.7	36.8	42.8	54.2	61.9	68.6	66.4	58.9	50.3	43.1	38.7		
		MCWB	34.0	33.0	33.7	38.3	47.5	55.1	62.9	61.0	55.6	47.3	41.6	37.1		
	0.4%	WB	38.4	38.6	41.2	45.5	56.5	62.2	70.2	69.6	60.3	52.9	46.4	41.9		
		MCDB	40.3	42.5	47.0	52.4	63.2	67.3	74.2	72.5	62.2	53.8	47.5	42.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	36.7	36.6	38.1	42.6	53.1	59.4	68.0	66.5	58.7	51.6	45.1	40.2		
		MCDB	38.0	39.2	42.6	47.6	59.4	64.5	72.3	70.0	60.6	52.2	46.2	41.2		
	5%	WB	35.4	34.6	35.8	40.6	50.5	57.6	65.9	64.0	57.4	49.8	43.6	38.7		
		MCDB	36.6	36.2	38.8	44.6	55.7	62.2	70.0	67.1	59.3	51.3	44.6	39.8		
	10%	WB	34.1	33.2	34.2	39.0	48.3	56.2	63.7	62.1	56.2	48.0	41.9	37.3		
		MCDB	35.3	34.5	36.3	41.9	53.1	60.5	67.4	65.2	58.1	49.8	42.9	38.5		
	5% DB	MDBR	8.9	10.6	11.1	10.1	10.9	10.6	9.8	10.1	9.5	8.1	6.7	8.3		
		MCDBR	7.7	9.6	13.0	15.0	15.1	14.1	12.9	12.0	11.1	8.3	5.9	7.0		
		MCWBR	7.1	7.6	8.7	9.4	8.5	8.2	7.6	7.7	7.0	5.4	6.5	6.5		
		MCWBR	7.3	9.1	11.8	14.2	14.3	11.5	10.8	10.4	8.5	6.8	5.7	6.8		
(35) Mean Daily Temperature Range	5% WB	MCWBR	6.8	7.6	8.4	9.4	8.8	7.7	7.1	7.3	6.9	6.3	5.5	6.5		
	taub		0.230	0.289	0.312	0.351	0.352	0.349	0.368	0.358	0.327	0.312	0.369	0.349		
		taud	2.051	2.188	2.260	2.326	2.451	2.496	2.466	2.489	2.553	2.412	2.672	2.606		
(42) Clear-Sky Solar Irradiance	Ebn at Noon		120	201	247	262	274	277	267	259	247	197	112	66		
		Edh at Noon	9	19	27	32	30	30	30	27	21	16	9	5		
	RadAvg		48	234	706	1194	1585	1800	1651	1252	757	303	77	23		
(44) All-Sky Solar Radiation	RadStd		6	45	79	112	170	122	161	108	81	55	16	2		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (45 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page